

Exam 1 Study Guide

September 2026

If I bold something in the slides it means it's important. I will NOT test on the required readings. I will do my best to avoid the situation where it's a lot of "Using your answer to part A, do part B". Expect a mix of short answer, multiple choice & true/false. There will be graphs. I would advise against memorizing the formulas (see section 10 for my comments on the math you need to know) and focusing on the larger swaths. Be comfortable calculating the five number summary and basic probability calculations.

1 Introduction

- When looking at a data set you can identify...
 1. what are the observations
 2. what are the variables
- Populations vs Samples
 - An observation is a single thing from a sample, a sample is a collection of things from the population, the population is all of the things
- Parameters vs Statistics
- Types of variables
 - Numeric
 - * Continuous
 - * Discrete
 - Categorical
 - * Nominal
 - * Ordinal

2 Data Visualization

- Why do we graph data?
- How do we choose which type of graph to use?

- Identify appropriate graphs for different types of data
- Describe the distribution for...
 - One categorical variable
 - * Using a bar chart mention
 - Proportion in each category
 - Something qualitative (Eg largest category is roughly x2 as large as the next biggest category)
 - One numeric variable
 - * Using a histogram be comfortable identifying...
 - the number of modes
 - symmetry vs skew
 - outliers
 - * Using a boxplot be comfortable identifying....
 - Symmetry vs skew
 - Outliers
 - Understanding a boxplot can't show modes
 - Two numeric variables
 - * Scatterplots
 1. Direction
 2. Strength
 3. Outliers
 4. Form
 - Numeric and Categorical
 - * Boxplots
 - Identify the 5 number summary using a boxplot
 - Building a boxplot from the 5 number summary
 - Discussing relative sizes of IQR between categories/boxplots

3 Numeric Summaries

- Drawbacks compared to data visualizations
- Moment Statistics
 - Mean
 - St. Dev. (formula not needed)
 - Assumptions (ie symmetric, no outliers in the distribution)
- Order Statistics
 - percentiles

- 5 number summary
- IQR ($Q_3 - Q_1$)
- What “robust” to outliers means
- Relative advantages and disadvantages between moment stats and order stats
- In what situations is the mean larger/smaller than the median (heavily skewed right/left respectively, or outliers in those directions)
- When to choose one over the other
- What do I mean when I talk about “processed” data?
- Conditional statistics
 - It’s just calculating a statistic(s) for a particular subpopulation
 - eg we find the distribution of income for only women

4 Correlation

- What does Pearson’s correlation coefficient, r , measure?
 - What’s the range of r ?
 - What’s the units of r ?
 - How, if at all, does r change if we convert our measurement units (eg mpg to kpl)
 - Does a large r imply a linear fit?
 - Type of data r can be used with?
- What is Spearman’s correlation?
 - What type of data can Spearman’s cor. be applied to?
 - When do we choose Spearman’s over Pearson’s?
 - What does a “monotonic” relationship mean
- Rank graphs by how strong their (Pearson) correlations are
- Understand that correlation can be either positive negative or zero, and what does having a negative correlation mean? 0 correlation?
- Ecological Fallacy (group aggregated statistics don’t apply on the individual level)
- Correlation vs Causation
- Lurking variable
 - Be prepared to give an example

5 Probability

- Define probability
- Empirical vs Theoretical vs Subjective probability
 - Empirical: Based off of data
 - * Law of Large Numbers: Empirical probability will approach the “true” probability
 - Theoretical: Based off of a mathematical model
 - Subject: Based off of vibe checks
- Identify a uniform distribution and use to find probabilities
 - Eg what’s the probability of rolling a 3 or 4 on a 4-sided dice (called a d4)?
- Use a uniform distribution to find a different distribution
 - Eg what is the probabilities associated with the sum of two d4’s? Ie if we roll two 4-sided dice and add them up what is the probabilities of the possible outcomes?
 - | Outcome | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|-------------|------|------|------|------|------|------|------|
| Probability | 1/16 | 2/16 | 3/16 | 4/16 | 3/16 | 2/16 | 1/16 |
- Marginal vs Conditional Probability
 - Be comfortable calculating these
 - Marginal: The probabilities of an event, ignoring other categories/variables
 - * Sampling 100 Americans (men and women) and reporting the proportions (probabilities) an American is in a political party
 - Conditional: The probabilities of an event within a subpopulation.
 - * Phrased as Variable A given Variable B
 - * Eg sampling 100 Americans (men and women) and reporting the proportions (probabilities) an American WOMAN is in a political party
 - * So “women” is our subpopulation and the variable we are looking at is the political party of the women

6 Math You’ll Need to be Comfortable With

- Calculating a five number summary
 - Min, Max are easy
 - Median is the middle number
 - Q_1 and Q_3 are the medians of the lower and upper halves of the data, respectively
- Calculating IQR given Q_1 and Q_3
- Probability
 - Conditional
 - Marginal